The Influence of Public Opinion on Policy Dynamics:
Exploring the determinants of change in European Policy-Making

by

Christine Arnold (University of Maastricht) and
Mark Franklin (European University Institute)

Paper prepared for delivery at the Workshop on
"The Determinants of Policy Change: Advancing the Debate"
European Consortium on Political Research
Helsinki, Finland
May 7-11, 2007

ABSTRACT

It has been pointed out many times that democratic governance requires a two-way flow of information. Governments must be able to react to what people want and people must be able to react to what governments do. Previous research on the responsive public has established a link between the state of domestic public opinion regarding Europe and the volume of supranational policy-making. This paper investigates the other half of the reciprocal link needed for thermostatic control of policy formation.

This paper examines the connection between public opinion and policy change asking, to what extent the level of public support and the degree of heterogeneity of public opinion can be utilized to explain the variance in the volume of European legislation both across policy fields and across time. Do the productions of the European Union reflect the wishes of the European public? In this way we plan to address the question to what extent European policymakers demonstrate awareness of heterogeneity and variability in public support for Europe.

The paper will employ data from Eurobarometer surveys conducted between 1971 and 2004. Additionally, it will use data from the Official Journal of the EU for the same time period to classify legislation into policy fields. We find evidence that policy change in the EU is influenced by public opinion. When voters want more policy in a certain area, policymakers do respond (after a lag of a year or so) and when they want less, policymakers respond to that also. The paper thus offers promise of helping explain variations in policy-making which otherwise are hard to account for.

Do the productions of the European Union reflect the wishes of the European public? Put another way, do those responsible for the mountains of legislation emanating from Brussels pay attention to the desires of those for whose ostensible benefit the legislation is enacted? The officials of the European Union are often accused, in the press and in casual conversation, of being insensitive to the wishes of ordinary Europeans, yet the Commission spends a fortune on evaluating and tracking the reactions of the European public to seemingly all things European, using six-monthly Eurobarometer polls among much else. If they did not wish to take account of public opinion, why would they do this?

It has been pointed out many times that democratic governance requires a two-way flow of information (see Deutsch 1966 for the classic exposition). Governments must be able to react to what people want and people must be able to react to what governments do. Early studies of public opinion and voting behavior called into question whether the average citizen would have such information (Berelson et al., 1954; Campbell et al., 1960; Converse, 1964, 1970). Analyses of American National Election Study panel data from 1956, 1958 and 1960 revealed that about half the respondents changed their opinions on policy issues as though they were answering at random. A general conclusion was that these people's policy attitudes as expressed in opinion polls were for the most part *nonattitudes* (Converse 1970). More recently, however, various scholars have proposed that even though many citizens may neither be very attentive to politics, nor very informed about all sorts of details, this does *not* mean that they are unable to vote rationally. With the aid of cognitive cues, they may still be able to evaluate candidates or parties in a meaningful way (Fiske and Linville, 1980; Feldman and Conover, 1983; Conover and Feldman, 1982, 1984; Popkin, 1991; Zaller, 1992; Sniderman, 1993; Kinder, 1993).

Building on this work, during the 1990s Christopher Wlezien (1995, 1996) tested the idea first propounded by Deutsch that the two-way flow of information can be regarded as a feedback loop in which the public performs like a thermostat, sending signals to the government to turn up the policy-making when they deem the rate of policymaking insufficient and to turn down the policy-making when its level exceeds what the public wants. Wlezien established that the thermostatic model of public opinion and policy worked well in certain policy areas (namely those areas that are salient to the public) but less well in other (less salient) areas (Wlezien 1995; 2004). The model has proved robust to changes in specification and data, being found to operate not only in the US but also in Canada and Britain (Soroka 2003; Wlezien and Soroka 2004, 2005, 2007).

The importance of issue salience as a requirement for public responsiveness to policy-making was established by using European integration policies as a test case, chosen because this was a policy area known to have increased in public salience over time (Franklin and Wlezien 1997). That research focused only on the public responsiveness portion of the feedback loop, since that was the information-flow that would demonstrate (or not) the importance of issue salience. In practice the European public performed as expected, reacting not at all to the productions of Brussels during the early years of the EEC/EC, but becoming quite responsive during the later period. Since the middle-1980s the European public has reacted with great sensitivity to outputs of European legislation from Brussels. It was not the purpose of that study to enquire into the other flow of information required for thermostatic governance – the flow from public to policy-makers – but Franklin and Wlezien noted in passing that European policy-makers did not appear to show much responsiveness to changes in the level of support for European policies among the European public.

The purpose of this paper is to take the obvious next step, and focus specifically on the extent to which European policymakers pay attention to European public opinion. To improve our chances of finding such attentiveness we look not only at overall EU policy-making, as studied by Franklin and Wlezien (1997), but also at policy-making in particular policy areas. We also look at support for European policies not only over Europe as a whole but also in individual countries (the twelve countries that have been members since 1985). Specifically we look at five policy areas about which we have public opinion data for all the member countries of the EC/EU going back to the start of opinion polling by means of bi-annual Eurobarometer studies. These are environmental policy, defense and foreign policy, welfare policy, regional development, and unemployment. These are areas in which EU legislation can be mapped onto public opinion regarding the policies concerned. We employ two questions asked in repeated Eurobarometer surveys, one regarding the salience of the policy area and one regarding the suitability of the area for EU policy-making. Responsive EU policy-makers would restrict their policy-making to areas which were deemed to be within their competence (not areas that people perceive as within the province of national governments) and which were deemed important. The extent of policy-making by the EU in each area is measured by the number of directives and regulations issued each year in the *Official Journal* of the EC/EU. We follow Franklin and Wlezien in ignoring other entries in the *Official Journal* as being more properly viewed as involving either preparation or implementation rather than as involving legislation (Thompson 1989:18).

Theory and operationalization

Wlezien's elegant thermostatic theory is normally expressed as a simple equation:

$$R_t = P'_t - P_t \qquad \text{Equation 1}$$

Where R_t is the public's relative preference for policy at time t (more or less), P'_t is their preferred level of policy at that time and P_t is the actual level of policy being produced. If P'_t is greater than P_t then the public's relative preference will be for more policy whereas if P'_t is less than P_t their relative preference will be for less policy. As pointed out by Wlezien (1995) this is precisely the mechanism that keeps temperature constant in a living space whose temperature is thermostatically controlled.

Two embellishments to this simple model are stressed by Wlezien. First, it should really incorporate an additional subscript, j, to indicate different policy areas. Second, it should be embellished with a measure of salience (S) which determines to what extent the public pays attention to the policy-area concerned.

Seen with those embellishments in terms of a conventional regression analysis, the equation would look like

$$R_{jt} = a_j + B_{1j}P'_{jt} + B_{2j}(P_{jt})(S_{jt}) + e_{jt} \qquad \text{Equation 2}$$

Note that we use beta coefficients in the equation so as to take account of variation in coefficients that reflect no more than differences in the metric of the variables (both within and across policy domains) which we expect to find in practice (see Wlezien 1995).

As already pointed out, our interest in this paper focuses more on policy-makers than on public opinion. Thus, after evaluating the above equation with our data for replication purposes, we will rearrange the coefficients to put policy on the left hand side, yielding an equation that looks like

$$P_{jt} = a_{j0} + B_{1j}(P'_{jt}) + B_{2j}(R_{jt})(S_{jt}) + e_{jt} \qquad \text{Equation 3}$$

In Equation 2 we expect the sign of the B_2 coefficient to be negative. In Equation 3, one of the coefficients on the right hand side must of necessity be negative, but we would expect it to the β_1 coefficient, for reasons that will be discussed later.

Operationalizing these equations with the data to hand is not totally straightforward. Wlezien (1995, 1996, 2004) measured policy outputs in terms of dollar expenditures, however it does not seem particularly problematic to regard them in terms of the volume of legislation (this is what, after all, is generally referred to by commentators: the sheer amount of regulation emanating from Brussels).

If we look at volume of EU legislation, we find that between 1952 and 2005 the EU passed a total of 105,989 primary and secondary legislative acts.[1] Primary legislation refers to the treaties of the European Union, while secondary legislation refers to regulations, directives, recommendations, and opinions issued by the Union's institutions in accordance with the treaties.[2] Since this paper is concerned with understanding the dynamics that drive the actual output of the legislation, only completed legislative acts by either the Council or the Commission are considered, and of these only directives and regulations are used in our models. Adding all regulations and directives that were passed in the EU since 1952 one arrives at a total of 62,025 pieces of legislation. Figure (1) shows the increase of EU legislation by type across the years. In this figure we can see that regulations make up the lion's share of EU legislation. Following the Treaty of Maastricht and Amsterdam a considerable jump in the number of regulations can clearly be observed.

[1] Not included in this number are preparatory acts such as proposals for legislation, written questions and communications. If these are included the total number comes to 246,291.
[2] This data was collected from Lexis Nexis which in turn draws on the data base Celex. It should be noted that the data points for 2005 terminate at the end of September.

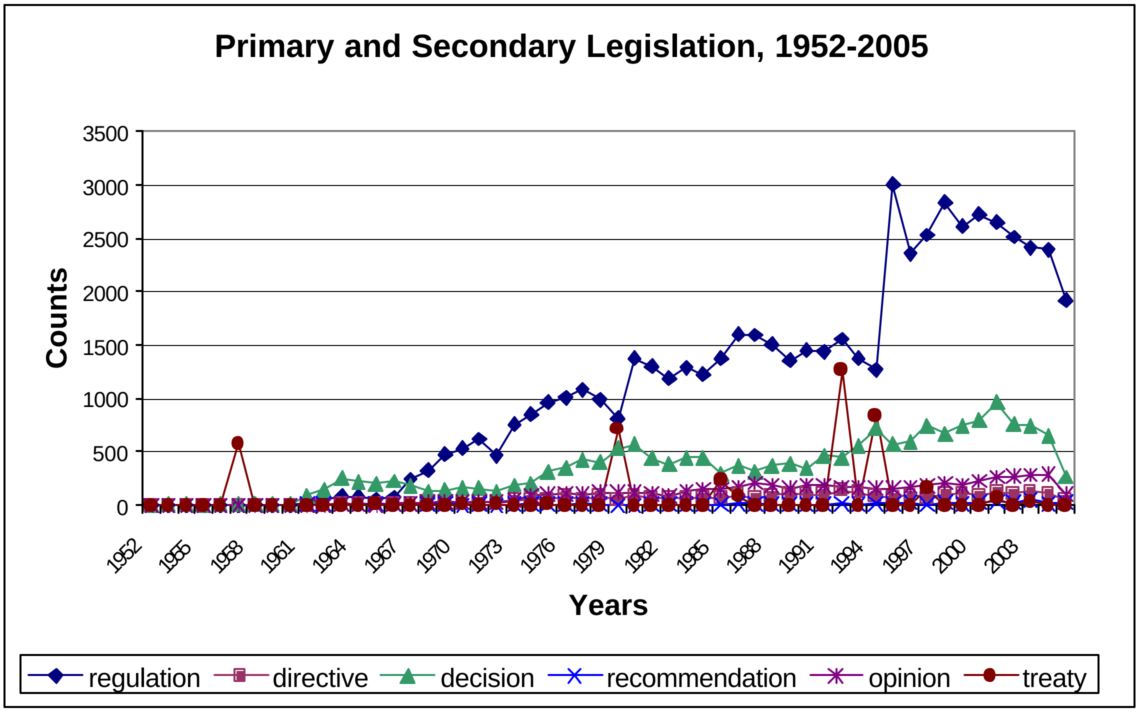

Figure 1 EU Legislation, 1952-2004

The unit selected to attribute each legislative act to a particular policy area is the "subject" field as provided by Lexis Nexis. It should be noted that a "subject" field was available for the vast majority of directives and regulations. However, it is important to point out that in about 30 percent of all cases no subject line was entered into the Lexis Nexis database (that is in 18,540 cases). In these cases, the keywords line was selected. This was done in 15,989 cases. Thus, by supplementing our analysis of the subject line with the keywords line, we attain a 96 percent completion rate for our coding of legislation into policy areas.

A dictionary was created to attribute to a policy area all terms from the subject line, or alternatively, if need be, all terms from the keywords line. The matching of terms to policy areas is done on the basis of information contained in the Eurovoc database.[3] Eurovoc is a multilingual thesaurus covering the policy fields in which the European Communities are active. The advantage of the Eurovoc thesaurus is that it provides a structured list of expressions intended to represent in an unambiguous fashion the conceptual content of

[3] The thesaurus can be accessed at the following website: http://europa.eu/eurovoc/

documents. As such it can provide a useful taxonomy for the semi-automatic classification of a large volume of documents. The quality of the thesaurus is assured by the fact that it appears to be extensively used by European and national institutions. As an indexing system it is used by the European Parliament, the Office for Official Publications of the European Communities, national and regional parliaments in Europe, and national governmental departments.

Figure (2) provides a visual representation of the number of directives and regulations differentiated by ten policy areas across time. In general the number of directives and regulations steadily increased since the late 1960s. Some policy areas had a peak between 1996 and 2003.

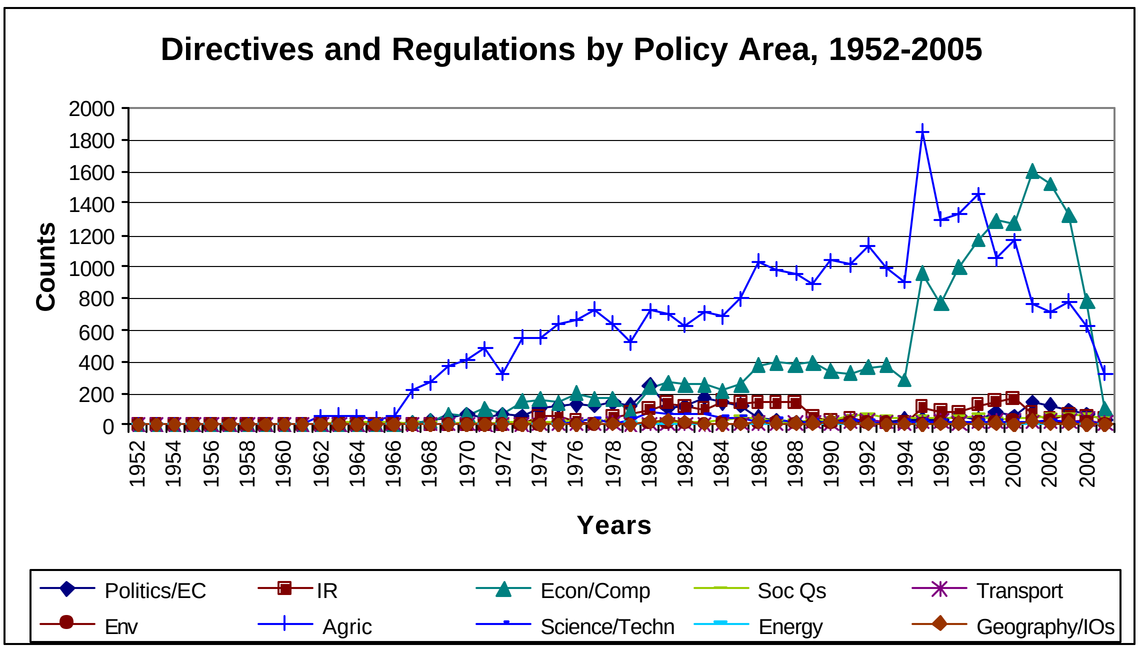

Figure 2 EU legislation by policy areas, 1952-2005

Turning to public opinion, Wlezien (1995, 1996, and 2004) focused on whether the public wanted the government to spend more, less, or about the same in specific policy areas. However, Franklin and Wlezien assumed that the amount of legislation wanted by the public would be a more appropriate counterpart to the volume of legislation produced. And Eurobarometer data provided variables that appeared to correspond to the preferred level of unification (membership in the EU a good thing) and relative preference for unification (in favor of unification). The first of these variables relates to a state of affairs (people either want unification or they do not) while the second relates to a process (people want unification either to move forward, move backward, or not move at all), as required for the thermostatic model. The unification question was removed from the EB questionnaire after 1995 (when the proportion of respondents wanting more unification dropped precipitously, suggesting a dramatic lowering of support for European unification), but we were able to extend the series by splicing it to another question, asked sporadically before 1995 but regularly thereafter, about the desired speed of European Integration. The variable in question (EUSPEED2) correlates 0.86 with the unification variable over the cases that the two variables have in common, and could be 'spliced' to the unification variable by lowering its mean and reducing its range. Though the rationale for these choices could be questioned, the proof of the pudding was in the eating. Using these variables the model performed as expected theoretically, with preferences for policy tracking changes in preferred level and changes in policy provision to an almost spectacular degree.

Franklin and Wlezien (1997) did not employ a measured indicator of the salience of European integration. They depended on the common perception that salience was increasing and used a trend variable that simply increased with the passage of time. One by-product of the present paper is a validation of Franklin and Wlezien's findings using a more appropriate measure of salience.

In regard to specific policy areas, the variables available to us in this research are a little different. Again we have two, one relating to a state of affairs (whether people think the EU should take common action in particular areas). The other, however, is not about process but about salience: how important people feel a particular common problem to be. This is good in that it provides us with a specific measure of salience, which was absent in the Franklin-Wlezien research, but it means we have no specific measure of whether people want more or less policy in these specific areas. The strategy we use to overcome this deficiency is to assume that any policy area in which common action is desired is a policy area in which more legislation is desired. We thus use a high value for the P' variable, arbitrarily set at 1 (which drops out of the analysis because it is constant). The rationale for this might be questionable in the case of a policy like regional development, in which the EU has been visibly involved for many years. Our other policy areas, however, are ones in which EU activity, if any, has received little publicity. Later we will use such expected differences between policy areas as a basis for testing theoretical expectations.

As in the case of the original research by Franklin and Wlezien, the proof of the pudding, regarding these operationalizations, will be in the eating. If we find meaningful relationships between these variables this will help to validate the way in which we have operationalized our theoretical expectations. If we do not reach significant findings, this will put our operationalizations into question. But there is another reason why our findings with regard to specific policy areas might fail to reach significance. This is that the questions regarding policy in specific areas were not asked in every year, and when questions about common action were asked they were not invariably matched by questions about common problems (the salience question). Rather than having data points for the large majority of years since 1970 (as is the case for the unification and membership questions), in specific policy areas our N is reduced to 12 or less (only 7 for regional development and the environment). In countries that only joined the EU in the middle 1980s, our time series is shorter and the Ns for these questions are sometimes too small for sensible analysis.

The responsiveness of public opinion to policy

We start by attempting to replicate the findings of Franklin and Wlezien (1997) over a longer time-period. Table 1, below, replicates their Table 1 which reported analyses of data for all of the (then) 12 EU member states taken together, over the years 1971 to 1994. In our Table 1 we bring the analysis more nearly up to date.[4] Our results, while having the same general import, are somewhat different in detail than those found for the shorter period. In particular, the effect of the trend variable is less than was found by Franklin and Wlezien. Using it as a surrogate for salience would not, over the longer period investigated here, give quite the same sense of the importance of increasing salience of the European issue over time. And using the trend variable squared has no effect at all on variance explained. These findings would not have provided the strong support found by Franklin and Wlezien for the

[4] In this preliminary report our time period extends only to 2001. The latest cumulative file of EB data goes to 2002; however, one of our critical variables (the spliced measure of desire for / speed of unification) is missing in that year.

critical influence of salience in conditioning the responsiveness of public opinion to policy change. Why is this?

Table 1. Support for unification regressions, 1971-2001 (standard errors in parenthesis)

Variable	Model A	Model B	Model C
Constant	0.445***	0.412***	0.404***
	(0.069)	(0.056)	(0.055)
Membership	0.527***	0.531***	0.525***
	(0.023)	(0.105)	(0.106)
Policy	-0.133**		
	(0.046)		
Policy * trend		-0.131***	
		(0.043)	
Policy * trend2			-0.145***
			(0.042)
Adjusted R-squared	0.626	0.670	0.670
N	26	26	26

***p<.001; **p<.01, *p<.05 (two-tailed)

We investigate the difference in findings by taking advantage of the fact that in our dataset we have a properly measured salience variable that was unavailable to Franklin and Wlezien.[5] This is the mean value of the common problem variables (how important each issue area was deemed to be) over all common problem questions asked in any year. When we substitute this variable for the trend variable in Table 1, we obtain the results shown in Models A and B of Table 2. Though the N is small, the results seem to perform exactly as expected theoretically. They show an undeniable influence of salience, which pushes the effects of policy over the threshold of statistical significance even with so small a number of cases, and raises variance explained considerably.

Having validated the new measure of salience, we can use it as a diagnostic to see why, over the longer period investigated in this paper, the trend measure failed to perform as

[5] The questions giving rise to this variable had not been asked sufficiently often, ten years ago, for the variable to prove viable in time series analysis. As we shall see, the series is still very sparse.

well as over the shorter period investigated by Franklin and Wlezien. Inspection of the data shows a clear ceiling effect for the salience measure, which ceases to rise after about 1993. This makes sense. The completion of the single market, the negotiations that gave rise to the Maastricht Treaty, and (above all) the much publicized difficulties found in ratifying the treaty in several countries, ensured that by the end of 1993 the issue of Europe had become fully salient. After that there are fluctuations in salience but no further sustained rise or fall.

With this in mind it is no longer reasonable to use as a surrogate for salience a trend measure that increases continuously. In order to replicate the ceiling effect shown by our true salience measure, we need to create a new trend variable that rises only until 1993 and is flat thereafter. Model C in Table 2 duplicates Model A in Table 1, the better to permit an appreciation of the influence of the revised trend variable. In Model D of Table 2 we see that the interaction with the new variable causes policy to achieve significance at the 0.001 level (for the first time in any of the models in Tables 1 or 2) and gives rise to a palpable increase in variance explained. The ability to validate the use of a surrogate measure of salience

Table 2 Support for unification, supplementary regressions, 1971-2001 (standard errors in parenthesis)

Variable	Model A	Model B	Model C	Model D
Constant	0.454 * (0.161)	0.448 ** (0.136)	0.445 *** (0.069)	0.443 *** (0.054)
Membership	0.534 (0.298)	0.520 (0.266)	0.527 *** (0.023)	0.544 *** (0.101)
Policy	-0.136 (0.072)		-0.133 ** (0.046)	
Policy * Salience		-0.157 * (0.065)		
Policy * New trend				-0.113 *** (0.030)
N	11	11	26	26
Adjusted R-squared	0.522	0.601	0.626	0.688

***p<.001; **p<.01, *p<.05 (two-tailed)

in this way is important for this research because our "proper" salience measure exists for less than half the dataset. Given the small number of cases at our disposal, it is useful to have a way of retaining all of them in any analysis involving salience.

Effects on policy-making

Having validated the most problematic of the measures at our disposal, we can now turn to the first of the main tasks before us in this paper, namely to discover whether policy-makers respond to public opinion in the same way as public opinion responds to policy. Put another way, we want to discover whether policy-makers in the EC/EU pay genuine attention to what the EU public wants.

Turning the responsiveness equation around, as in Equation 3 above, raises tricky issues of causal priority. It is logically possible for the European public to become aware of policy enactments in the very year of the enactments concerned, because promulgation follows a legislative process that is more or less public, even in the EC/EU. It is presumably that process that enables (some) members of the European public to become aware of the legislation concerned, rather than the actual publication of the resulting laws in the *Official Journal*. So, when Franklin and Wlezien (1997) established that public opinion responds to policy-making in the same year as the policies were made, this did not stretch credulity.[6] Much the same immediacy of response to policy-making has been established in the United States and elsewhere (Wlezien 1995; Wlezien and Soroka 2007). However, when we turn the equation around, and try to predict policy from public opinion, the same sort of immediacy is not to be expected. It takes time to bring policies to fruition and it would be quite implausible to imagine that policy-makers could on a regular basis react so fast to changes in public opinion as to be able as a matter of course to enact and publish legislation in the same year as the need for that legislation became apparent. So causal impact on policy-making must of necessity arise from public desires or needs expressed in the previous year or earlier. The policy model must, to be plausible, incorporate significant lagged effects.

[6] We find in our data that, although the strongest impact on current opinion cones from current legislation, legislation in the previous year does have almost as much impact. The two variables (present year legislation and previous year legislation) are so strongly related, however, that it is not possible to include both of them in the same prediction equation.

We begin, in Table 3, by considering various models involving the same three variables as were investigated in Tables 1 and 2. Model A serves the same purpose as Model A in Table 1 (and Model C in Table 2) – a benchmark for what happens when we do not take increasing salience of European unification into account. It is no surprise that one of the coefficients is negative. When we rearrange the components of Equation 2 to make Equation 3, moving policy to the left hand side necessarily gives it a positive sign, and the rules of algebra ensure that one of the other variables has to acquire a negative sign. The surprise is that it is the unification variable that acquires the negative sign in Model A. Clearly this is totally unreasonable. It implies that more European policies are produced in defiance of an expressed desire for less policy. Taking increasing salience of unification policies into account, in Model B, causes the negative coefficient to move onto the membership variable. This might be thought equally implausible, but if anything is clear from these models it is that policymakers cannot be taking both expressed evaluations of membership (good or bad) and expressed desire for unification (more or less) into account. If they take account of

Table 3 Policy responsiveness to public opinion in multivariate analysis, with various lags in the measurement of independent variables, 1971-2001

Variable	Model A Lag=0		Model B Lag=0		Model C Lag=1	``	Model D Lag=2	
Constant	1.595 (0.295)	***	0.785 (0.149)	***	0.799 (0.146)	***	0.883 (0.133)	***
Membership	0.395 (0.603)		-1.411 (0.314)	***	-1.406 (0.308)	***	-1.481 (0.288)	***
Unification	-1.962 (0.689)	**						
Unification * New trend			1.028 (0.172)	***	1.038 (0.167)	***	0.981 (0.158)	***
N	26		26		25		24	
Adjusted R-squared	0.296		0.628		0.642		0.661	

***p<.001; **p<.01, *p<.05 (two-tailed)

either of these measures they must ignore the other, and it makes sense that they should ignore the measure that does not overtly involve policymaking. We will evaluate this assumption in due course. But first we should note that when a one-year lag for the independent variables is introduced (in Model C), both the effect of unification and the total variance explained goes up. Past public opinion evidently has a heavier hand in conditioning policy-making (if the effects are real) than past policy-making has in conditioning public opinion (see footnote 4). This is as it should be if we are to find plausible a claim that opinions about policy-making are taken into account by policy-makers, as discussed above. When considering opinions two-years earlier (Model D) we find a somewhat lower effect of opinion about unification, but a somewhat higher R^2. Beyond two years previously, policy preferences have both smaller effects and give rise to lower R^2 (though neither of the two preference variables completely loses significance until the opinions concerned have passed more than five years into the past).

Table 4 Policy responsiveness to public opinion, opinion variables taken separately, with various lags: 1971-2001

Years past	Membership b	Unification * New trend b
Contemporaneous	-0.765 (0.483)	0.803 *** (0.191)
Lag=1	-0.751 (0.479)	0.798 *** (0.188)
Lag=2	-0.835 (0.469)	0.738 *** (0.187)
Lag=3	-0.699 (0.473)	0.766 *** (0.183)

***p<.001; **p<.01, *p<.05 (two-tailed)

The idea that policy-makers do not take public opinion regarding membership into account but do take note of opinion regarding unification becomes more plausible when we look at the results of taking each of these variables separately as the sole predictor of the volume of policy-making. In Table 4 it can be seen that, whether contemporaneously or up to three years previously, opinions on membership never significantly affect the volume of policy-making (the standard error is almost small enough when Lag=2, but then rises again when Lag=3). The unification variable – actually the splice of the unification (more or less) with the unification (faster or slower) variables – is, by contrast, highly significant both contemporaneously and lagged up to three years in the past. On the basis of these findings it seems plausible to suppose that policy-makers do take account of expressed wishes for more or less unification when deciding how much policy to produce. In the next sections of this paper we will try to validate this conjecture by looking separately at individual policy areas and at individual countries.

Heterogeneity in responses to policy preferences

If policy-makers take account of public opinion, we would expect them to do so regarding public opinion in each country individually as well as regarding all countries taken together. The alternative – that policy-makers in Brussels favor some countries over others in terms of following the wishes of the citizens of particular countries – would surely not be something that they could permit themselves to do on any regular basis.[7] By contrast, if policy-makers are indeed following public opinion when deciding how much policy to enact in different areas, then we would expect to find differences between policy areas depending on the nature of those areas. Thus a first source of support for the conjecture that policy-makers take account of public opinion when deciding how much policy to enact in different areas would be to find homogeneity across countries but heterogeneity across policy areas in terms of responsiveness to public support for unification policies. Establishing whether this is indeed the case is the subject of this section.

In order to investigate differences across countries and policy areas we need to disaggregate our data to the country-year level, multiplying our case base almost twelve-fold to yield some 300 units - enough to enable us to make distinctions between countries and policy areas. For each country-year we have opinion data at the appropriate level of

[7] Of course this assumption might be wrong, in which case (unless it is wrong in a readily comprehensible fashion) a finding of heterogeneity among countries, in terms of the extent to which policy-makers follow the policy preferences of citizens in different countries, will falsely lead us to question our conjecture. But, if homogeneity is found in the effects across countries, then we can be more confident that our conjecture is true.

aggregation, with the unification variable and variables regarding common problems and common actions aggregated to the country-level rather than being averaged across all countries (as was done for the analysis presented earlier in this paper). These data still emply the same measure of policy output, however, as was employed in previous analyses (total of regulations and directives, overall or in particular policy areas). This gives us a statistical problem when calculating significance levels, because the policy variable is highly clustered by year. In what follows we thus calculate robust standard errors clustered by year.

Table 5 reports on an analysis by country in which policy responsiveness is judged, as in Table 4, by a lagged (Lag=2) version of the interaction between policy salience (indicated by the new trend variable) and support for (more) unification (spliced to the speed up, slow down variable). The table contains a row for each country, and we can clearly see that our expectation that the lagged opinion variable would have a significant effect on policy-making is supported in all but four countries. These are Britain, Denmark, Portugal, and Spain. Not only are effects for the other countries always statistically significant, their magnitudes are quite homogeneous also, with none of the significant effects being as much as twice as strong as any other.

Table 5 Policy responsiveness to average desire for more unification by country (Lag = 2), 1971-2004 (standard errors in parentheses)

Country	b	s.e.		R^2	N
Belgium	0.678	(0.227)	**	0.241	26
Britain	0.430	(0.418)		0.002	25
Denmark	0.555	(0.324)		0.072	25
France	0.572	(0.253)	*	0.142	26
Germany	0.962	(0.217)	***	0.437	26
Greece	0.647	(0.155)	***	0.447	22
Ireland	0.751	(0.172)	***	0.431	25
Italy	0.611	(0.151)	***	0.379	26
Luxembourg	1.092	(0.296)	***	0.344	25
Netherlands	0.886	(0.253)	**	0.311	26
Portugal	0.385	(0.290)		0.046	17
Spain	0.573	(0.565)		0.002	17

***p<.001; **p<.01, *p<.05 (two-tailed)

Portugal and Spain have a shorter time series available for analysis than other countries, which might explain the lack of significant effects. But it is also possible that EU policymakers pay less attention to pubic opinion in countries that have been members of the EU for only a relatively short time. Britain and Denmark are also countries that were not founder-members of the EU, which lends support to this idea. But Ireland joined the EC at the same time as Britain and Denmark, and Irish public opinion does appear to have a significant effect on EU policy-making. So perhaps public opinion in Britain and Denmark fails to affect EC/EU policy-making because these are known Euroskeptic countries whose citizens enjoy less influence on that account.

Another possible reason for the disparities seen in Table 5 would be that EU policy-makers cannot follow minority opinions. If public opinion in Britain, Denmark, Spain and Portugal takes positions which are minority positions in other EU countries, then policy-makers would need to ignore the opinions of the majorities in those countries. The exact reason for the disparities we observe will need to be investigated in future research. For the moment we note that these particular four countries do have distinguishing characteristics that might well set them apart from mainstream EU countries in terms of opinions regarding Europe and in terms of the weight that EU policymakers might give to opinions in those countries. So, while the letter of our expectations regarding homogeneity of effects on policymaking by country are not met, still we should not too easily reject the evidence of homogeneity in terms of the support it brings to our conjecture about the responsiveness of European policy-makers.

If, instead of disaggregating by country we disaggregate our data by policy area, we find considerably more heterogeneity. Only 3 of the 5 policy areas see significant effects, and the difference between the largest and smallest of these effects is almost five-fold.[8] Welfare and regional development policy areas do not see significant effects of public opinion on policy-making, whereas environment, defense, and unemployment are areas in which effects are significant. Even in these areas, however, there is considerable diversity in the magnitudes of effects. These differences, like those between countries, make sense. The EC/EU has been deeply and visibly involved in promoting the development of backward

[8] The unification variable is the spliced more/less with faster/slower , as in Table 5. It is interacted with the new trend variable (as in Tables 2-5). The alternative interaction with measured salience yields even greater heterogeneity, with the largest effect being almost 7 times as great as the smallest.

regions and in facilitating the delivery of welfare benefits to citizens residing in countries other than those where the benefits were earned. It is reasonable to suppose that some citizens would not want the EU to further extend its remit in these areas even though they believe the areas to be important and agree with the role the EU has taken in these areas. Defense and unemployment, on the other hand, are areas in which the EU has taken few well-publicized initiatives. It is reasonable that those who think they are areas in which the EU should be involved would want the EU to take action. It is also reasonable to suppose that environmental policies would fall somewhere between these extremes.

Table 6 Policy responsiveness to average desire for more unification by policy area (lag=2), 1971-2004 (standard errors in parenthases)

Policy area	b	s.e.		R^2	N
Environment	0.892	(0.396)	*	0.038	103
Defense	2.060	(0.165)	***	0.457	186
Welfare	0.649	(0.532)		0.007	67
Regional development	0.480	(0.724)		---	83
Unemployment	0.429	(0.082)	***	0.132	176

***p<.001; **p<.01, *p<.05 (two-tailed)

So the heterogeneity of policy-making responsiveness in different policy areas makes it appear that EU policy-makers may indeed be paying attention to public opinion, reinforcing the message given by the homogeneity of responsiveness when this is broken down by country.

Discussion

This paper represents an initial cut at evaluating the extent to which policy-making in the EC/EU responds to public opinion. We find that the volume of policy emanating from Brussels seems to respond to desires for more or less (faster or slower) unification expressed over the previous two or three years by EU citizens. This responsiveness matches the responsiveness of public opinion to policies enacted in the same year, already established in past research (Franklin and Wlezien 1997) and replicated over a longer period in this paper.

Causation is always difficult to prove, but the difference in timing between the reactions of the public and the reactions of policy-makers adds plausibility to our conjecture that policy-makers do respond to expressed desires for more or less legislation, as does the heterogeneity of responsiveness within different policy areas in contrast to the homogeneity found when investigating the impact of opinion in different countries (at least as far as founder-members of the EC/EU are concerned).

These initial findings raise almost as many questions as they settle. Future research will need to extend the number of policy areas investigated and come up with a better understanding of why some policy-areas see more responsiveness on the part of policymakers than other policy areas do. Perhaps examining a larger number of policy areas from the perspective of the EU pillars, might bring into perspective an underlying pattern. Future research also needs to address the question of why public opinion in certain countries appears to have more influence on policy-makers than public opinion in other countries. Though it seems not unreasonable that opinions expressed by citizens of founder-member countries should have more weight than opinions expressed by other citizens, the exact reasons for the differences need to be investigated. Possibly, drawing on taxonomy of institutions of political representation from comparative politics might facilitate a better explanation of the observed variance. One additional promising line of investigation might be to tap into implementation and compliance research from EU studies. This might help to gain a better understanding of the degree to which EU directives are actually implemented in a timely fashion at the national level and the variation in the number of subnational actors involved in the transposition process.

These additional investigations may serve to buttress the findings reported in this paper and increase the plausibility of the conjecture on which it is based. The idea that thermostatic governance is possible even where elections do not serve the functions expected of them in democratic societies (van der Eijk and Franklin 1996; Franklin 2005) is a fascinating one, and mapping out the mechanisms involved may yield valuable insights not only about the governance of the EC/EU but also about governance within nation states.

REFERENCES

Berelson, B.R., P.F. Lazarsfeld, and W.N. McPhee. 1954. Voting. *A Study of Opinion Formation in a Presidential Campaign.* University of Chicago press, Chicago.

Campbell, A., P.E. Converse, W.E. Miller, and D.E. Stokes. 1960. *The American Voter*. Wiley, New York.

Conover, P.J., and S. Feldman. 1984. "How People Organize the Political World: A Schematic Model." *American Journal of Political Science* 28:95-126.

Converse, Philip. 1964. "The nature of belief systems in mass publics." In: D. Apter (ed.), *Ideology and discontent.* New York: Free Press.

Converse, Philip. 1970. "Attitudes and Non-attitudes: Continuation of a Dialogue," In: E.R. Tufte (ed.), *The Quantitative Analysis of Social Problems.* Reading: Addison-Wesley..

Feldman, S., and P.J. Conover.1983. "Candidates, Issues, and Voters: the Role of Inference in Political Perception," *Journal of Politics* 45:812-39.

Fiske, S.T., and P.W. Linville. 1980. "What Does the Schema Concept Buy Us?" *Personality and Social Psychology Bulletin* 6:543-57.

Franklin, Mark, and Christopher Wlezien. 1997. "The Responsive Public: Issue Salience, Policy Change, and Preferences for European Unification," *Journal of Theoretical Politics* 9, 347-63.

Karl Deutsch. 1966. *The Nerves of Government*. New York: Free Press.

Kinder, Donald. 1993. "Coming to Grips with the Holy Ghost," In: D.R. Kinder, and T.R. Palfrey (eds.). *Experimental Foundations of Political Science.* University of Michigan Press, Ann Arbor.

Popkin, S. 1991. *The Reasoning Voter.* University of Chicago Press, Chicago.

Sniderman, Paul. 1993. "The New Look in Public Opinion Research," In: A.W. Finifter (ed.), *Political Science: The State of the Discipline II.* The American Political Science Association, Washington.

Soroka, Stuart. 2003. *Agenda-Setting Dynamics in Canada.* Vancouver: University of British Columbia Press.

Soroka, Stuart and Christopher Wlezien. 2004. "Opinion Representation and Policy

Feedback: Canada in Comparative Perspective," *Canadian Journal of Political Science*, vol. 37

Thompson, Ian. 1989. *The Documentation of the European Communities: A Guide.* London: Mansell.

van der Eijk, Cees and Mark Franklin. 1996. *Choosing Europe? The European Electoral and National Politics in the Face of Union.* Ann Arbor, MI: University of Michigan Press.

Wlezien, Christopher. 1995. "The Public as Thermostat: Dynamics of Preferences for Spending," *American Journal of Political Science* 39:981-1000.

Wlezien, Christopher. 1996. "Dynamics of Representation: The Case of U.S. Spending o Defense," *British Journal of Political Science* 26:81-103.

Wlezien, Christopher. 2004. "Patterns of Representation: Dynamics of Public Preferences and Policy," *Journal of Politics*, 66.

Zaller, John. 1992. *The nature and origins of mass opinion.* Cambridge: Cambridge University Press.